

A Point-Like Dark-Matter Target at 12.8 GHz from a QCD-Anchored Vacuum Condensate: Theory and Haloscope Test Protocol

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Abstract

The Null-Vector Gravity (NVG) framework, anchored to a single QCD input $M_{\Omega,0} = 859 \pm 8$ MeV fixed by lattice sigma terms, contains a pseudo-Goldstone mode θ of the vacuum condensate $\Phi = W e^{i\theta}$. We examine the phenomenological consequences of the hypothesis that the same decay constant f_a that sets the θ -mode mass also generates Majorana neutrino masses through a Weinberg-like dimension-five operator suppressed by a two-loop QCD-anomaly coefficient. Introducing the unknown overall coefficient of this operator as an explicit parameter c_ν , the benchmark choice $c_\nu = 1$ fixes $f_a = 1.07 \times 10^{11}$ GeV from the neutrino sector and yields a *conditional point target* for dark matter: a coherent pseudo-scalar of mass $m_\theta = 53.2$ μ eV, detectable as a microwave line at $f_\theta = 12.86$ GHz with photon coupling $g_{\theta\gamma\gamma} = 2.08 \times 10^{-14}$ GeV $^{-1}$ (KSVZ benchmark). The product $m_\theta f_a$ agrees with the QCD axion band at 2.6% — a normalization cross-check, not an independent test — and the benchmark frequency lies in the currently unexplored gap between the published ORGAN mass windows (25–26 and 63–67 μ eV), directly on the lower edge of the SMASH band (50–200 μ eV) that ORGAN has declared as its roadmap target. We present the derivation, a calibration of the haloscope signal formula against the published ORGAN signal estimate, a Monte Carlo propagation of the statistical input uncertainties (m_3, χ_{top}), together with explicit conditional sensitivity scenarios for the unknown coefficient c_ν , and two concrete test protocols: (A) a resonant-cavity array and (B) a *fixed-position* dielectric booster stack, where the point target eliminates the scanning mechanism. For a coherent dark-matter fraction of 24% (misalignment only), protocol B reaches SNR = 5 in ~ 18 h at power boost $\beta^2 = 10^4$ (MADMAX convention, $\beta^2 \equiv P_{\text{sig}}/P_0$). The target is extended to nucleon, electron and neutrino channels with no additional free parameters, and a minimal UV completion of the θ -seesaw is sketched in Appendix 8. Discovery and exclusion criteria are preregistered. This is a *phenomenological target paper*: the θ -seesaw operator is stated as a hypothesis, its coefficient is propagated as a parameter, and every derived quantity is conditional on it. All numerical values are reproducible from the accompanying scripts.

Keywords: axion dark matter, haloscope, dielectric haloscope, pseudo-Goldstone boson, vacuum condensate, neutrino mass, phenomenological target

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1 Introduction

Direct searches for QCD axion dark matter have so far produced exclusion limits in disjoint mass windows: ADMX covers $\sim 2.7\text{--}4.2\ \mu\text{eV}$ [14], HAYSTAC covers $\sim 17\text{--}20\ \mu\text{eV}$ [15], ORGAN has scanned $25.45\text{--}26.27\ \mu\text{eV}$ [18], $63.2\text{--}67.1\ \mu\text{eV}$ [16] and $107.42\text{--}111.93\ \mu\text{eV}$ [17], while MADMAX is under construction for the $40\text{--}400\ \mu\text{eV}$ range [21]. The intermediate region around $50\text{--}60\ \mu\text{eV}$ ($12\text{--}15\ \text{GHz}$) remains unexplored.

This note reports that the NVG/VMF framework [2], anchored to a single QCD input [1], yields — conditional on one stated hypothesis (the θ -seesaw, Sec. 2.3) — a point target *exactly* in that gap: a dark-matter pseudo-scalar at $53.2\ \mu\text{eV}$. Because the target is a *point* in (m, g_γ) space rather than a band, the experimental problem simplifies drastically: no scanning, no tuning mechanism — a fixed-frequency integration.

Section 2 gives the derivation. Section 3 states the observable signature. Section 5 documents the gap in published searches. Section 6 defines the test protocol, including a calibration of the signal formula against published data and preregistered discovery/exclusion criteria. Section 4 extends the target to nucleon, electron and neutrino channels; Appendix 8 sketches a minimal UV completion of the θ -seesaw. Section 7 lists the framework assumptions honestly.

2 Theoretical Framework

2.1 The anchor: $M_{\Omega,0}$ from lattice sigma terms

The framework is anchored to one dimensionful QCD input. Separating the nucleon mass into current-quark and nonperturbative components via the Feynman–Hellmann sigma terms ($\sigma_{\pi N} \approx 44\ \text{MeV}$, $\sigma_{sN} \approx 30\ \text{MeV}$), the vacuum value of the nonperturbative component is fixed at [1]

$$M_{\Omega,0} = 859 \pm 8\ \text{MeV}, \tag{1}$$

which sets the condensate energy density scale $\rho_c \approx 7.09 \times 10^4 \text{ MeV/fm}^3$ used throughout the construction. No cosmological or astrophysical input is used at any stage.

2.2 The θ pseudo-Goldstone mode

The NVG vacuum condensate is parametrized as $\Phi = W e^{i\theta}$, where the phase θ is a pseudo-Goldstone mode with the standard anomaly-generated potential. Its mass obeys the same relation as the QCD axion,

$$m_\theta = \frac{\sqrt{\chi_{\text{top}}}}{f_a}, \quad (2)$$

with the topological susceptibility $\chi_{\text{top}}^{1/4} = 75.5 \text{ MeV}$ from lattice QCD [10]. In the normalization $m_a(f_a = 10^{12} \text{ GeV}) = 5.691 \text{ } \mu\text{eV}$ [10], this becomes $m_\theta = 5.691 \text{ } \mu\text{eV} \times (10^{12} \text{ GeV}/f_a)$.

2.3 The θ -seesaw: fixing f_a from the neutrino sector

The framework hypothesis behind the target is the θ -seesaw: the same mode θ generates Majorana neutrino masses without right-handed neutrinos. To be consistent with the electroweak gauge group above EWSB, the operator must have the unique gauge-invariant dimension-five (Weinberg) structure [7]; the NVG hypothesis specifies only its suppression scale and coefficient,

$$\mathcal{L}_{\text{eff}} = 2 c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{1}{f_a} (\tilde{H}^\dagger L)^T C (\tilde{H}^\dagger L) + \text{h.c.}, \quad \tilde{H} \equiv i\sigma_2 H^*, \quad (3)$$

where c_ν is an unknown dimensionless coefficient. The normalization convention of this Weinberg-like operator, including symmetry factors in flavor space and the factors of $1/\sqrt{2}$ from $\langle H \rangle = v_{\text{EW}}/\sqrt{2}$, is absorbed into c_ν ; accordingly, $c_\nu = 1$ defines a benchmark normalization rather than a UV-derived coefficient. The operator is *defined* through the physical post-EWSB relation, for the heaviest neutrino,

$$m_3 = c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{v_{\text{EW}}^2}{f_a}. \quad (4)$$

With $\alpha_s(M_Z) = 0.1184$, $v_{\text{EW}} = 246.22 \text{ GeV}$ and the oscillation-and-cosmology anchor $m_3 = 50.3 \text{ meV}$ (normal ordering; the implied sum $\sum m_\nu = 59 \text{ meV}$ passes the DESI DR2 bound $\sum m_\nu < 64 \text{ meV}$ unconditionally), Eq. (4) fixes

$$\boxed{f_a = c_\nu \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{v_{\text{EW}}^2}{m_3} = 1.07 \times 10^{11} c_\nu \text{ GeV}} \quad (5)$$

We stress the status of Eq. (3): it is a *framework hypothesis*, the single non-standard ingredient of this note. The nominal coefficient $(\alpha_s/4\pi)^2 \approx 8.9 \times 10^{-5}$ is motivated by the two-loop QCD-anomaly diagram, but possible flavor factors and RG running are not computed here, and the UV completion that generates the operator is not derived. The parameter c_ν absorbs all of this freedom: the benchmark $c_\nu = 1$ defines the headline target, and every result below is rescaled with c_ν (Section 6.4). The haloscope test discriminates the hypothesis directly (Section 7).

2.4 Point target for the dark-matter mass

Because every derived quantity depends on c_ν through f_a alone, the parametric structure is transparent (for fixed E/N):

$$f_a = f_{a,0} c_\nu, \quad m_\theta = \frac{m_{\theta,0}}{c_\nu}, \quad f_\theta = \frac{f_{\theta,0}}{c_\nu}, \quad g_{\theta\gamma\gamma} = \frac{g_{\theta\gamma\gamma,0}}{c_\nu}, \quad (6)$$

where the subscript 0 denotes the benchmark values at $c_\nu = 1$. A larger c_ν means a larger f_a , hence a *smaller* mass and frequency and a weaker photon coupling. Inserting (5) into (2):

$$m_\theta = 5.691 \text{ } \mu\text{eV} \times \frac{10^{12}}{1.07 \times 10^{11} c_\nu} = \frac{53.2}{c_\nu} \text{ } \mu\text{eV} \quad (7)$$

or, equivalently, a Compton frequency

$$f_\theta = \frac{m_\theta c^2}{h} = \frac{12.86}{c_\nu} \text{ GHz}. \quad (8)$$

The benchmark $c_\nu = 1$ gives $m_\theta = 53.2 \text{ } \mu\text{eV}$, $f_\theta = 12.86 \text{ GHz}$ throughout the rest of this note.

Point versus band. If a UV completion fixes c_ν , the target is a genuine *conditional point* in (m, g_γ, f) space. If c_ν remains free, the point degrades to a *broad conditional search band* along the one-parameter curve (6) — 4.3–38.6 GHz for $c_\nu \in [1/3, 3]$ (Sec. 6.4) — and the fixed-frequency advantage is retained only conditionally, at a chosen benchmark value of c_ν . We use the terminology “conditional point target” and “broad conditional search band” accordingly.

Normalization cross-check (not an independent test). Once Eq. (2) is accepted, the product $m_\theta f_a = \sqrt{\chi_{\text{top}}}$ is fixed by the same QCD input, so its landing on the QCD axion band $m_a f_a \simeq m_\pi f_\pi \sqrt{z}/(1+z) \approx 5.82 \times 10^{15} \text{ eV}^2$ [10] cannot count as an independent confirmation of NVG or of the seesaw hypothesis. The framework gives $m_\theta f_a = 5.67 \times 10^{15} \text{ eV}^2$: the 2.6% deviation is a numerical consistency check of two lattice normalizations (the χ_{top} of [10] against the VMF anchor), and nothing more. The genuinely independent content of the target is the *value* of f_a itself, fixed by the neutrino sector.

2.5 Coupling to photons

The effective interaction is the standard axion-electrodynamics term [8]

$$\mathcal{L} \supset -\frac{g_{\theta\gamma\gamma}}{4} \theta F_{\mu\nu} \tilde{F}^{\mu\nu} = g_{\theta\gamma\gamma} \theta \mathbf{E} \cdot \mathbf{B}, \quad g_{\theta\gamma\gamma} = \frac{\alpha}{2\pi} \frac{|E/N - 1.92|}{f_a}. \quad (9)$$

For the KSVZ benchmark [5] ($E/N = 0$) and the DFSZ benchmark [6] ($E/N = 8/3$):

$$g_{\theta\gamma\gamma}^{\text{KSVZ}} = 2.08 \times 10^{-14} \text{ GeV}^{-1}, \quad g_{\theta\gamma\gamma}^{\text{DFSZ}} = 8.10 \times 10^{-15} \text{ GeV}^{-1}. \quad (10)$$

The model-dependent factor $|E/N - 1.92|$ is the only residual theory uncertainty on the coupling. We emphasize that $g_{\theta\gamma\gamma}$ is *not predicted* by NVG: the framework fixes f_a but not the UV anomaly ratio E/N , so KSVZ and DFSZ are benchmarks, not consequences of the model. A null result at KSVZ level therefore excludes only KSVZ-like realizations (Section 6.4). The KSVZ value is used as the discovery benchmark.

2.6 Relic abundance: the honest coherent fraction

For a post-inflationary scenario with the natural initial misalignment $\theta_i = \pi/\sqrt{3}$, the standard misalignment estimate [9] gives

$$\Omega_\theta h^2 \simeq 0.12 \theta_i^2 \left(\frac{f_a}{10^{12} \text{ GeV}} \right)^{7/6} = 0.0291, \quad (11)$$

i.e. **24% of the local dark-matter density** ($\Omega_{\text{DM}} h^2 = 0.120$). The remaining density may reside in topological defects of the condensate sector; it does not contribute to the coherent haloscope signal. All signal powers below use the conservative coherent fraction $f_{\text{coh}} = 0.24$; the $f_{\text{coh}} = 1$ values are also quoted where relevant. A larger coherent fraction (e.g. from defect decay) would strengthen the signal proportionally.

2.7 Fifth force: the pseudoscalar null channel

The Compton range $\lambda_\theta = \hbar c/m_\theta = 3.72$ mm falls inside the sub-millimetre window of torsion-balance experiments [12]. Because θ is a *pseudoscalar*, the leading-order monopole–monopole Yukawa exchange between unmixed states vanishes, so unpolarized Eöt-Wash-class searches are expected to see nothing at this range. The residual channels — spin-dipole exchange with coupling m_N/f_a , loop-induced scalar admixtures, two-boson exchange, or condensate mixing — require a dedicated analysis and are not claimed here as quantitative predictions; we state only the robust null expectation for the unpolarized monopole channel.

3 Experimental Signature

The benchmark target is summarized in Table 1.

Table 1: Benchmark target of the NVG θ -DM sector, conditional on the θ -seesaw hypothesis with $c_\nu = 1$. No quantity in the right column is fitted to haloscope data.

Quantity	Predicted value	Origin
Decay constant f_a	1.07×10^{11} GeV	neutrino sector, Eq. (4)
Mass m_θ	53.2 μ eV	Eq. (2), lattice χ_{top}
Frequency f_θ	12.86 GHz	$m_\theta c^2/h$
Virial linewidth Q_a	$\sim 10^6$	halo velocity dispersion
Line width Δf_a	12.8 kHz	f_θ/Q_a
$g_{\theta\gamma\gamma}$ (KSVZ)	2.08×10^{-14} GeV $^{-1}$	Eq. (9)
$g_{\theta\gamma\gamma}$ (DFSZ)	8.10×10^{-15} GeV $^{-1}$	Eq. (9)
Coherent DM fraction	24%	misalignment, Sec. 2.6
$m_\theta f_a$ vs QCD band	−2.6% deviation	consistency check

The signal is a nearly monochromatic microwave excess at 12.86 GHz with $Q_a \sim 10^6$ intrinsic quality factor, linearly polarized along the applied magnetic field, scaling as B_0^2 , and tracking the sidereal modulation of the halo wind (a secondary discriminant).

4 Multi-Channel Interaction Block

The photon coupling of Sec. 2 is one entry of the full low-energy interaction block. Completing it makes the target multi-channel falsifiable without introducing new free parameters: the fermion axial charges below are fixed either by the same UV benchmark choice (KSVZ-like or DFSZ-like) already made for E/N , or — in the neutrino channel — by the θ -seesaw operator itself.

4.1 Effective Lagrangian and channel benchmarks

$$\mathcal{L}_{\text{int}} = -\frac{g_{\theta\gamma\gamma}}{4} \theta F \tilde{F} + \frac{\partial_\mu \theta}{2f_a} \sum_f C_f \bar{f} \gamma^\mu \gamma^5 f, \quad (12)$$

with $f = e, p, n, \nu_3$. Using the fermion equations of motion, each derivative coupling is equivalent to a pseudoscalar Yukawa coupling $g_{\theta ff} = C_f m_f/f_a$.

Nucleons. The axial charges split into a model-independent chiral piece (axion–pion mixing, fixed by QCD) and the UV charges C_q^0 [27]:

$$C_p = -0.47 + 0.88 C_u^0 - 0.39 C_d^0 - 0.49 C_s^0, \quad C_n = -0.02 - 0.02 C_u^0 + 0.88 C_d^0 - 0.49 C_s^0. \quad (13)$$

For the KSVZ-like benchmark ($C_q^0 = 0$ — the same UV choice as $E/N = 0$), $C_p = -0.47$ and $C_n = -0.02$ are *predictions*, not parameters: $g_{\theta pp} = -4.1 \times 10^{-12}$ and $g_{\theta nn} = -1.8 \times 10^{-13}$. The spin-dependent fifth-force channel of Sec. 3 is thereby upgraded from the order-of-magnitude quote m_N/f_a to a definite coupling.

Electrons. The two photon benchmarks separate sharply here: DFSZ-like UV gives a tree-level $C_e = \cos^2\beta/3$ (illustratively 0.30 at $\tan\beta = 3$, $g_{\theta ee} \simeq 1.4 \times 10^{-15}$), while KSVZ-like UV gives $C_e = 0$ at tree level and a loop-suppressed $C_e \sim 10^{-5}$ ($g_{\theta ee} \sim 5 \times 10^{-20}$). A 4.5-order difference makes the electron channel a UV discriminator. Both values pass the white-dwarf cooling constraint $g_{ae} \lesssim 3 \times 10^{-13}$ [12] with wide margins.

Neutrinos. The θ -seesaw operator itself carries the PQ phase, so the mass term reads $m_3 e^{i\theta/f_a}$: the heaviest neutrino obtains an axial charge $C_{\nu_3} = 1$ in the normalization of Eq. (3), i.e. $g_{\theta\nu_3\nu_3} \simeq m_3/f_a = 4.7 \times 10^{-22}$. This coupling is a distinctive NVG signature — it is absent in plain KSVZ and DFSZ — and the decay $\theta \rightarrow \nu\nu$ is kinematically forbidden ($2m_3 \gg m_\theta$), so the channel affects stability of the dark-matter candidate not at all.

Table 2: Multi-channel benchmark couplings at $f_a = 1.07 \times 10^{11}$ GeV ($c_\nu = 1$), all scaling as $1/c_\nu$. Rows marked “benchmark” inherit the same UV choice as the photon channel; the neutrino row and the flavor scenario of Sec. 4.2 follow from the θ -seesaw itself.

Channel	KSVZ-like benchmark	DFSZ-like ($\tan\beta = 3$, illustrative)
Photon $g_{\theta\gamma\gamma}$ (GeV $^{-1}$)	2.1×10^{-14}	8.1×10^{-15}
Nucleon $g_{\theta pp}$	-4.1×10^{-12}	$\sim -2 \times 10^{-12}$
Nucleon $g_{\theta nn}$	-1.8×10^{-13}	$\sim -1 \times 10^{-13}$
Electron $g_{\theta ee}$	$\sim 5 \times 10^{-20}$ (loop)	1.4×10^{-15} (tree)
Neutrino $g_{\theta\nu_3\nu_3}$	4.7×10^{-22} (NVG-specific)	

4.2 Rank-one flavor scenario

If the Weinberg-like operator is rank one and aligned with the heaviest mass eigenstate — as the single-coefficient chain (4) assumes — normal ordering with $m_1 = 0$ is the natural case. With the NuFIT 5.2 oscillation parameters [30], this *illustrative scenario* yields the correlated triple

$$\sum m_\nu = 58.9 \text{ meV}, \quad m_\beta = 8.8 \text{ meV}, \quad m_{\beta\beta} \in [1.5, 3.7] \text{ meV}, \quad (14)$$

where the $0\nu\beta\beta$ value carries a *nonzero floor* ($|U_{e2}^2 m_2 - U_{e3}^2 m_3|$ at destructive Majorana phase). The inverted-ordering alternative gives $\sum m_\nu \approx 106$ meV, so the rank-one hypothesis is a clean cosmological discriminator: it places the mass sum at the normal-ordering minimum, and a measurement $\sum m_\nu \gtrsim 75$ meV (CMB-S4/DESI-class sensitivity) would exclude it. As with the c_ν scenarios of Sec. 6.4, rank-one alignment is a scenario assumption, not a framework derivation.

4.3 Experimental mapping

The predicted nucleon couplings fall inside the design window of NMR/spin-precession programs: the CASPER design band $f_a \sim 10^9$ – 10^{12} GeV [29] contains $f_a = 1.07 \times 10^{11}$ GeV, and at $m_\theta = 53.2 \mu\text{eV}$ the oscillating nuclear EDM channel (CASPER-electric type) and the spin-gradient channel (CASPER-wind, QUAX, ARIADNE type) address the same mass with a *predicted* coupling rather than a scanned one. The electron channel is a UV discriminator rather than a prediction; the fifth-force channel of Sec. 3 remains qualitative pending a computation of the loop-induced scalar admixture.

5 Status of Published Searches

Table 3 compiles the published haloscope coverage around the predicted mass.

Table 3: Published haloscope mass windows around the NVG target. The benchmark point (53.2 μeV) lies in the unexplored gap between the two ORGAN windows, at the lower edge of the SMASH band (50–200 μeV) [11] that ORGAN declares as its roadmap target [16].

Experiment	Mass window (μeV)	Frequency (GHz)	Best limit on $g_{a\gamma\gamma}$ (GeV^{-1})
ADMX [14]	2.66–3.31	0.64–0.80	KSVZ reached
HAYSTAC II [15]	~ 17.3 –19.5	4.18–4.71	$\sim 10^{-13}$
ORGAN [18]	25.45–26.27	6.15–6.35	2.8×10^{-13}
NVG target	53.2	12.86	2.08×10^{-14}
ORGAN [16]	63.2–67.1	15.3–16.2	3.0×10^{-12}
ORGAN 1b [17]	107.42–111.93	26.0–27.1	cogenesis ALP excluded
MADMAX [21, 24]	40–400 (design)	10–100	prototype phase

Two remarks. First, no published limit reaches $g = 2.1 \times 10^{-14} \text{ GeV}^{-1}$ anywhere near 53 μeV : the closest result (ORGAN at 25.5–26.3 μeV) is a factor ~ 13 weaker in coupling, at a different mass. Second, the ORGAN collaboration’s stated goal is to scan the SMASH band 50–200 μeV under a constant magnetic field [16]; the NVG benchmark sits exactly at the lower edge of that band, making a targeted single-frequency run a natural addition to their program.

6 Test Protocol

6.1 Signal-power formula and its calibration

For a microwave cavity haloscope [8, 25],

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \frac{\rho_\theta}{m_\theta} B_0^2 V C \min(Q_L, Q_a), \quad (15)$$

where $\rho_\theta = f_{\text{coh}} \rho_{\text{DM}}$ with $\rho_{\text{DM}} = 0.4 \text{ GeV}/\text{cm}^3$, V is the cavity volume, C the mode form factor, and Q_L the loaded quality factor.

Calibration against published data. The ORGAN collaboration quotes, for its Phase 1a apparatus, $P_{\text{sig}} \simeq 2.3 \times 10^{-25} \text{ W}$ (KSVZ) and $3.1 \times 10^{-26} \text{ W}$ (DFSZ) for a 64 μeV axion [16]. Two checks:

1. *Apparatus-independent*: the coupling-coefficient ratio $(g_{\text{KSVZ}}/g_{\text{DFSZ}})^2 = (0.97/0.36)^2 = 7.26$ versus the published power ratio $2.3/0.31 = 7.42$ — agreement at the 2% level, confirming the normalization convention of Eq. (9).
2. *Absolute*: back-solving Eq. (15) from the published $2.3 \times 10^{-25} \text{ W}$ implies an apparatus product $VCQ \approx 0.44 \text{ m}^3$ (for $B_0 = 7 \text{ T}$), consistent with an ORGAN-scale cavity at 15.6 GHz with $Q \sim 10^4$ – 10^5 . The unit chain of the present calculation is thus validated against a real haloscope at the factor-of-two level.

For the dielectric booster concept [19, 20, 21] the analogous expression is

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \rho_\theta B_0^2 \frac{A \beta^2}{m_\theta^2}, \quad (16)$$

where A is the disk area and $\beta^2 \equiv P_{\text{sig}}/P_0$ is the *power* boost relative to a bare magnetized mirror ($P_0 = g_{\theta\gamma\gamma}^2 \rho_\theta B_0^2 A/m_\theta^2$), following the MADMAX convention [24] in which β itself denotes

the amplitude boost; all boost figures quoted in this note are power boosts β^2 . Both Eq. (15) and (16) have been re-derived within the Lorentz-reciprocity framework [25, 26]; an in-situ reciprocity measurement with a source antenna is the final normalization check of the stack before any dark-matter run.

6.2 Protocol A: resonant cavity

At $f_\theta = 12.86$ GHz the TM_{010} mode requires a cylinder of radius $R = x_{01}c/(2\pi f_\theta) = 8.92$ mm. With length $L = 50$ mm, $V = 12.5 \text{ cm}^3$, form factor $C = 0.69$, $B_0 = 10$ T and a superconducting (Nb) cavity with baseline $Q_L = 10^5$,

$$P_{\text{sig}} = 6.3 \times 10^{-25} \text{ W} \quad (f_{\text{coh}} = 0.24). \quad (17)$$

A single cavity reaches $\text{SNR} = 5$ only after ~ 48 years: it is a pathfinder, not the discovery configuration. The viable cavity route is a phase-locked array (Table 4); $Q_L = 10^6$ at 13 GHz is flagged optimistic (Nb BCS losses scale as ω^2), so the $Q = 10^5$ rows are the engineering baseline.

Table 4: Integration time to $\text{SNR} = 5$ for cavity arrays ($B_0 = 10$ T, $T_{\text{sys}} = 1$ K, $f_{\text{coh}} = 0.24$, per-cavity signal from Eq. (15)).

	$N = 64$	$N = 256$
$Q_L = 10^5$ (baseline)	276 days	69 days
$Q_L = 10^6$ (optimistic)	0.3 days	0.1 days

6.3 Protocol B: fixed-position dielectric booster stack

The dielectric haloscope [19] replaces the resonant cavity by ~ 80 dielectric disks (LaAlO_3 , $\varepsilon \approx 24$) in a strong magnetic field; constructive interference of the emitted photons provides the power boost β^2 . In a scanning experiment the disk positions must be moved mechanically — the technically hardest subsystem, now being commissioned [23]. **The point-frequency target eliminates this subsystem entirely:** at a known frequency the inter-disk spacing is fixed, and the stack becomes a static optical element.

With $D = 1.2$ m disks ($A = 1.13 \text{ m}^2$, the low-mass-end MADMAX aperture), $B_0 = 10$ T and $f_{\text{coh}} = 0.24$, Eq. (16) gives (Table 5). The power-boost targets $\beta^2 = 10^4$ – 10^5 are the design goals of the MADMAX program [21]; the three-disk prototype has so far demonstrated β^2 up to ~ 2000 with 13–17% uncertainty [24], which we state as the principal technical risk of this option.

Table 5: Fixed dielectric stack, $B_0 = 10$ T, $T_{\text{sys}} = 1$ K, readout bandwidth equal to the axion linewidth $\Delta f_a = 12.8$ kHz (the stack boost band $\gg \Delta f_a$).

Power boost β^2	$P_{\text{sig}} (f_{\text{coh}} = 0.24)$	$P_{\text{sig}} (f_{\text{coh}} = 1)$	$t(\text{SNR} = 5)$
10^4	$3.1 \times 10^{-23} \text{ W}$	$1.3 \times 10^{-22} \text{ W}$	18 h
10^5	$3.1 \times 10^{-22} \text{ W}$	$1.3 \times 10^{-21} \text{ W}$	11 min

Protocol B is strictly simpler than the cavity array: no cryogenic tuning mechanism, no phase-locked multi-cavity readout, a single wide-bore magnet, horn antenna and a near-quantum-limited amplifier — while the required power boost β^2 is already the stated MADMAX design target in a frequency band that includes 12.86 GHz.

6.4 Readout, statistics, and preregistered criteria

Noise. $T_{\text{sys}} \approx 1$ K is adopted: millikelvin physical temperature plus a flux-driven Josephson parametric amplifier. This is not an assumption but demonstrated technology — ORGAN’s most recent phase operates exactly this way [18].

Statistical target definition. With the Dicke radiometer relation

$$\text{SNR} = \frac{P_{\text{sig}}}{k_B T_{\text{sys}}} \sqrt{\frac{t}{\Delta f}}, \quad (18)$$

the narrow search window below is conditional on the benchmark coefficient $c_\nu = 1$. Statistical uncertainties in m_3 and χ_{top} are propagated by Monte Carlo (Appendix 8: $m_3 \sim \mathcal{N}(50.3, 0.25)$ meV, $\chi_{\text{top}}^{1/4} \sim \mathcal{N}(75.5, 0.7\%)$ MeV), while c_ν is *not* assigned a statistical prior: it is an explicit theory parameter. Broader intervals obtained by varying c_ν below are sensitivity scenarios, not confidence intervals.

- **Conditional benchmark window ($c_\nu = 1$ fixed).** The target-frequency distribution has median 12.86 GHz and a 68% interval [12.67, 13.05] GHz (relative width 1.5%, dominated by χ_{top}). The corresponding pre-specified 3σ analysis window is [12.29, 13.43] GHz, resolved in bins of width Δf_a .
- **Illustrative coefficient scenario (c_ν varied over $[1/2, 2]$).** Via Eq. (6) this non-statistical sensitivity scenario broadens the target range to 6.4–25.7 GHz. It is a scan, not a point-target test, and must carry its full trials factor and sensitivity variation across the band.
- **Extreme envelope ($c_\nu \in [1/3, 3]$).** $f_\theta \in [4.3, 38.6]$ GHz (~ 9 times the benchmark band): the full scan cost if c_ν is never fixed by theory.

Signal power and sensitivity across the band. Along the curve (6), *at fixed coherent fraction*, the signal powers scale simply: $P_{\text{sig}}^{\text{stack}} \propto g^2/m_\theta^2 \propto c_\nu^0$ for the dielectric stack (exactly invariant under c_ν rescaling) and $P_{\text{sig}}^{\text{cavity}} \propto g^2/m_\theta \propto c_\nu^{-1}$ for the cavity, so the integration times of Tables 4–5 apply unchanged to the stack and rescale as c_ν^2 for the cavity at any other benchmark value of c_ν . If instead the misalignment estimate of Sec. 2.6 is imposed, the coherent fraction itself scales as $f_{\text{coh}} \propto f_a^{7/6} \propto c_\nu^{7/6}$, which multiplies both powers accordingly. Across coefficient scenarios the required sensitivity is therefore not constant: at fixed E/N , $g_{\theta\gamma\gamma} \propto c_\nu^{-1}$, while the assumed misalignment abundance also changes with $f_a \propto c_\nu$; consequently, a broad- c_ν program requires a frequency-dependent sensitivity forecast rather than a uniform scan-time rescaling. The $\pm 0.5\%$ window of the v1 draft covered only m_3 and χ_{top} and is superseded.

Trials factor and global significance. The benchmark prediction is a *conditional narrow target* in model-parameter space, but the experimentally searched frequency interval still contains many statistically independent axion-line templates: the 1.14 GHz benchmark window holds $N_{\text{eff}} \simeq 8.9 \times 10^4$ independent bins of width $\Delta f_a = 12.8$ kHz. A local 5σ excess scanned over the whole window corresponds to only $\sim 2\sigma$ globally, so the discovery statistic must report both a local significance and a global significance corrected for the pre-specified N_{eff} . Pre-registration fixes the analysis window and procedure; it does not remove the trials correction.

Discovery criterion. A candidate must exceed a pre-specified local threshold corresponding to a *global* 5σ false-alarm probability after correction for N_{eff} templates in the benchmark window. For the conservative independent-bin estimate above, $N_{\text{eff}} \simeq 8.9 \times 10^4$ implies a local threshold of approximately 6.9σ . The final N_{eff} shall be obtained from the matched-filter bank

and verified with noise-only Monte Carlo. The candidate must also persist under independent re-taking, scale as B_0^2 when the magnet current is varied, match the expected Maxwell–Boltzmann halo lineshape, and be reproduced at a second apparatus with a different magnet before any claim.

Exclusion criterion. Exclusion is reported at two levels. (1) A null result at KSVZ-level sensitivity, $g_{\theta\gamma\gamma} \leq 2.1 \times 10^{-14} \text{ GeV}^{-1}$ across the preregistered window, excludes the *KSVZ-benchmark realization* of the θ -seesaw target at $f_{\text{coh}} = 0.24$ — and only that realization, since E/N is UV-dependent. (2) A null at DFSZ level ($8.1 \times 10^{-15} \text{ GeV}^{-1}$) additionally covers the DFSZ benchmark; excluding the neutrino–axion linkage itself would further require a fixed E/N assignment and a cosmological determination of f_{coh} . Because exclusion curves assume the full local density, the effective bound on g for a coherent fraction f_{coh} is weaker by $1/\sqrt{f_{\text{coh}}} \approx 2.0$; this rescaling must be applied when interpreting any null.

Side-channel cross-checks. The same f_a predicts $\sum m_\nu = 59 \text{ meV}$ (CMB+BAO, ongoing) and, in the general case, a $0\nu\beta\beta$ window $m_{\beta\beta} \in [0, 6.4] \text{ meV}$, narrowing to the nonzero range $[1.5, 3.7] \text{ meV}$ under the rank-one scenario of Sec. 4.2 (LEGEND-1000, nEXO): these provide framework-level corroboration independent of haloscopes.

7 Honest Caveats

1. **Framework hypothesis and its coefficient.** The θ -seesaw, Eq. (3), is a postulate of the NVG construction. The coefficient c_ν — encoding the two-loop anomaly diagram, possible flavor factors and RG running — is unknown and is propagated as a parameter throughout; $c_\nu = 1$ is a benchmark, not a derived value. Appendix 8 sketches a minimal UV completion and shows that the benchmark corresponds to a perturbative, τ -Yukawa-scale neutrino coupling with ~ 5 –10% RG running — i.e. it is natural — but it does not fix c_ν . If Eq. (4) fails — e.g. m_3 from a future $0\nu\beta\beta$ /oscillation combination disagrees with 50.3 meV — the entire frequency target moves. The haloscope test is precisely the discriminator.
2. **Spectral premise.** The input $m_3 = 50.3 \text{ meV}$ assumes normal ordering and a near-minimal lightest neutrino mass. Oscillation experiments measure only mass-squared differences, and cosmology bounds the sum of masses without fixing m_3 at this precision. The target is therefore conditional on this premise; the Monte Carlo window of Sec. 6.4 covers the stated uncertainty, and a shift to inverted ordering or a heavier lightest state would move the target outside the benchmark window.
3. **Model dependence of the coupling.** The factor $|E/N - 1.92|$ in Eq. (9) is benchmarked to KSVZ/DFSZ; a different UV completion rescales $g_{\theta\gamma\gamma}$ without moving the frequency. The mass target is unaffected; the coupling is not predicted.
4. **Coherent fraction.** All powers use $f_{\text{coh}} = 0.24$ (misalignment only). If most of the sector’s density sits in incoherent defects, haloscope signals weaken accordingly; if defects decay into coherent modes, they strengthen. This is the dominant astrophysical uncertainty and is why the exclusion criterion carries the $1/\sqrt{f_{\text{coh}}}$ rescaling.
5. **Technology risk.** $Q_L = 10^6$ at 13 GHz and $\beta^2 = 10^5$ are program goals, not demonstrated hardware (the MADMAX prototype reached $\beta^2 \sim 2000$). The baseline rows ($Q_L = 10^5$, $\beta^2 = 10^4$) reflect the validated state of the art.

8 Conclusion

This note is a phenomenological target paper. Under the stated θ -seesaw hypothesis with benchmark coefficient $c_\nu = 1$, a single neutrino-sector input of the NVG/VMF framework fixes $f_a = 1.07 \times 10^{11}$ GeV, which through lattice-QCD physics yields a conditional point target: $m_\theta = 53.2$ μ eV, a microwave line at 12.86 GHz with KSVZ-level photon coupling. The point lands on the QCD axion band within 2.6% (a normalization cross-check, not an independent test), sits in the currently unexplored gap between published ORGAN windows, and is directly on that collaboration’s declared SMASH roadmap. Because the target is a point, the search needs no scanning: a fixed dielectric booster stack reaches discovery-level significance in under a day of integration at design power boost $\beta^2 = 10^4$. The protocol above is preregistered, the frequency window is derived by Monte Carlo from all stated uncertainties, and the exclusion statements are scoped precisely: a KSVZ-level null closes the KSVZ-benchmark realization of the target, not every conceivable θ -model. Either outcome is informative — which is the property a target must have to deserve an experiment.

Appendix A: Reproducibility

All numbers in this note are produced by scripts in the accompanying repository (<https://github.com/infosave2007/vmf>, Zenodo-archived releases):

- `generator_project/calc_theta_haloscope_design.py` (v4) — geometry, signal powers (Eqs. (15)–(16), MADMAX β^2 convention), ORGAN calibration, integration times, published-search status, the c_ν -parametrized θ -seesaw chain, the Monte Carlo block that produces the preregistered window of Sec. 6.4, the trials-factor estimate of the discovery criterion, and the multi-channel interaction block (nucleon/electron/neutrino couplings, rank-one flavor scenario, UV matching) of Sec. 4 and Appendix 8;
- `generator_project/calc_theta_haloscope_v2.py` — frequency derivation, axion-band consistency check (2.6%);
- `verification/nvg_neutrino_seesaw.py` — Eq. (4), fixing f_a ; `verification/nvg_theta_fifth_force.py` — pseudoscalar null channel.

Appendix B: A Minimal UV Completion of the θ -Seesaw

This appendix shows that the benchmark coefficient $c_\nu = 1$ is *natural* — generated by perturbative, non-hierarchical UV physics — and quantifies the RG running absorbed into c_ν . It does *not* predict c_ν : the coefficient remains a free parameter of the framework.

Field content and PQ charges. A global $U(1)_{\text{PQ}}$ is spontaneously broken by a scalar σ (charge $X_\sigma = 2$) at the scale f_a ; the phase of σ is θ . The spectrum beyond the Standard Model is:

- one right-handed neutrino N ($X_N = 1$), with Majorana mass from $\frac{1}{2}y_\sigma\sigma N^T C N$: $M_N = y_\sigma f_a/\sqrt{2} \equiv \kappa f_a$, carrying the PQ phase $e^{i\theta/f_a}$;
- lepton doublets and charged-lepton singlets with $X_L = X_{e_R} = 1$, so that both $\bar{L}\tilde{H}N$ and $\bar{L}He_R$ are PQ-invariant while quarks and H are neutral;
- one vectorlike color-triplet Q (KSVZ-like, electromagnetically neutral), which makes $U(1)_{\text{PQ}}$ anomalous under color and yields $E/N = 0$ — the photon benchmark of Sec. 2.

Generation of the Weinberg operator. Integrating out N at tree level through the standard seesaw diagram $L\tilde{H}-N-\tilde{H}^\dagger L^\dagger$ produces

$$\mathcal{L}_W = \frac{y_\nu^2}{2M_N} \frac{\sigma^\dagger}{|\sigma|} (\tilde{H}^\dagger L)^T C (\tilde{H}^\dagger L) + \text{h.c.}, \quad (19)$$

i.e. the Weinberg operator of Eq. (3) with its PQ phase $e^{-i\theta/f_a}$ generated automatically. Matching to the normalization of Eq. (4) gives

$$c_\nu = \frac{y_\nu^2}{2\kappa(\alpha_s/4\pi)^2} = 5.6 \times 10^3 \frac{y_\nu^2}{\kappa}. \quad (20)$$

What the matching does — and does not — imply. The benchmark $c_\nu = 1$ with $\kappa = 1$ corresponds to $y_\nu = 1.3 \times 10^{-2} \approx 1.8 y_\tau$: a perturbative Yukawa of τ -lepton magnitude, with no hierarchy and no fine-tuning. The scenario range $c_\nu \in [1/3, 3]$ corresponds to $y_\nu \in [7.7 \times 10^{-3}, 2.3 \times 10^{-2}]/\sqrt{\kappa}$, i.e. a factor of ~ 2 around y_τ . Perturbativity ($y_\nu^2 \lesssim 4\pi$) alone bounds $c_\nu \lesssim 7 \times 10^4/\kappa$, which is not restrictive: absent additional flavor dynamics, c_ν is not calculable, and the haloscope test of Sec. 6 remains the discriminator.

RG running. The one-loop running of the Weinberg coefficient in the Standard Model [28],

$$16\pi^2 \frac{dC_W}{d\ln\mu} = (-3g_2^2 + 6y_t^2 + \lambda)C_W + (Y_e^\dagger Y_e)C_W + C_W(Y_e^\dagger Y_e)^T, \quad (21)$$

shifts the coefficient by ~ 5 –10% between $M_N = f_a$ and the electroweak scale; this is absorbed into c_ν and is subleading relative to the benchmark scenario granularity. The two-loop QCD-anomaly suppression $(\alpha_s/4\pi)^2$ in Eq. (3) refers to the mechanism that ties the operator's normalization to the QCD scale; Eq. (20) shows how any given UV realization maps onto the effective coefficient.

Neutrino coupling and domain walls. The phase $e^{i\theta/f_a}$ in the mass term yields the axial coupling $C_{\nu_3} = 1$ ($g_{\theta\nu_3\nu_3} \simeq m_3/f_a$) used in Sec. 4 — the NVG-specific signature absent in plain KSVZ/DFSZ. The colored sector carries unit anomaly, so the domain-wall number is $N_{\text{DW}} = 1$ and any walls formed in the post-inflationary scenario are unstable, consistent with the cosmology of Sec. 2.6.

References

- [1] O. Kirichenko, *Lattice Sigma Terms as an Anchor for the Dense Nuclear Matter Equation of State*, Zenodo preprint, doi:10.5281/zenodo.20214457 (2025).
- [2] O. Kirichenko, *NVG-Research: Null-Vector Gravity and Vacuum Mass Fraction framework, verification suite and scripts*, <https://github.com/infosave2007/vmf>.
- [3] S. Weinberg, *A New Light Boson?*, Phys. Rev. Lett. **40**, 223 (1978).
- [4] F. Wilczek, *Problem of Strong P and T Invariance in the Presence of Instantons*, Phys. Rev. Lett. **40**, 279 (1978).
- [5] J. E. Kim, *Weak-Interaction Singlet and Strong CP Invariance*, Phys. Rev. Lett. **43**, 103 (1979); M. A. Shifman, A. I. Vainshtein, V. I. Zakharov, *Can Confinement Ensure Natural CP Invariance of Strong Interactions?*, Nucl. Phys. B **166**, 493 (1980).

- [6] A. R. Zhitnitsky, *On Possible Suppression of the Axion Hadron Interactions*, Sov. J. Nucl. Phys. **31**, 260 (1980); M. Dine, W. Fischler, M. Srednicki, *A Simple Solution to the Strong CP Problem with a Harmless Axion*, Phys. Lett. B **104**, 199 (1981).
- [7] S. Weinberg, *Baryon- and Lepton-Nonconserving Processes*, Phys. Rev. Lett. **43**, 1566 (1979).
- [8] P. Sikivie, *Experimental Tests of the Invisible Axion*, Phys. Rev. Lett. **51**, 1415 (1983).
- [9] J. Preskill, M. B. Wise, F. Wilczek, *Cosmology of the Invisible Axion*, Phys. Lett. B **120**, 127 (1983); L. F. Abbott, P. Sikivie, *A Cosmological Bound on the Invisible Axion*, Phys. Lett. B **120**, 133 (1983); M. Dine, W. Fischler, *The Not-So-Harmless Axion*, Phys. Lett. B **120**, 137 (1983).
- [10] G. Grilli di Cortona, E. Hardy, J. Pardo Vega, G. Villadoro, *The QCD axion, precisely*, JHEP **01** (2016) 034, arXiv:1511.02867.
- [11] G. Ballesteros, J. Redondo, A. Ringwald, C. Tamarit, *Standard Model-Axion-Seesaw-Higgs Portal Inflation. Five problems of particle physics and cosmology solved in one stroke*, arXiv:1610.01639 (2016).
- [12] I. G. Irastorza, J. Redondo, *New experimental approaches in the search for axion-like particles*, Prog. Part. Nucl. Phys. **102**, 89 (2018).
- [13] P. W. Graham, I. G. Irastorza, S. K. Lamoreaux, A. Lindner, K. A. van Bibber, *Experimental Searches for the Axion and Axion-Like Particles*, Annu. Rev. Nucl. Part. Sci. **65**, 485 (2015).
- [14] N. Du et al. (ADMX Collaboration), *A Search for Invisible Axion Dark Matter with the Axion Dark Matter Experiment*, Phys. Rev. Lett. **120**, 151301 (2018), arXiv:1804.05750.
- [15] X. Bai et al. (HAYSTAC Collaboration), *Dark Matter Axion Search with HAYSTAC Phase II*, Phys. Rev. Lett. **134**, 151006 (2025), arXiv:2409.08998.
- [16] A. Quiskamp et al. (ORGAN Collaboration), *Direct search for dark matter axions excluding ALPogenesis in the 63- to 67- μ eV range with the ORGAN experiment*, Sci. Adv. **8**, eabq3765 (2022).
- [17] A. Quiskamp et al. (ORGAN Collaboration), *Exclusion of Axionlike-Particle Cogenesis Dark Matter in a Mass Window above 100 μ eV*, Phys. Rev. Lett. **132**, 031601 (2024), arXiv:2310.00904.
- [18] A. P. Quiskamp et al. (ORGAN Collaboration), *Near-quantum-limited axion dark matter search with the ORGAN experiment around 26 μ eV*, Phys. Rev. D **111**, 095007 (2025), arXiv:2407.18586.
- [19] A. Caldwell et al. (MADMAX Working Group), *Dielectric Haloscopes: A New Way to Detect Axion Dark Matter*, Phys. Rev. Lett. **118**, 091801 (2017), arXiv:1611.05865.
- [20] A. J. Millar, G. G. Raffelt, J. Redondo, F. D. Steffen, *Dielectric Haloscopes to Search for Axion Dark Matter: Theoretical Foundations*, JCAP **01** (2017) 061, arXiv:1612.07057.
- [21] P. Brun et al. (MADMAX Collaboration), *A new experimental approach to probe QCD axion dark matter in the mass range above 40 μ eV*, Eur. Phys. J. C **79**, 186 (2019), arXiv:1901.07401.
- [22] S. Knirck et al. (MADMAX Collaboration), *Simulating MADMAX in 3D: Requirements for Dielectric Axion Haloscopes*, JCAP **10** (2021) 034, arXiv:2104.06553.

- [23] B. Ary Dos Santos Garcia et al. (MADMAX Collaboration), *First mechanical realization of a tunable dielectric haloscope for the MADMAX axion search experiment*, JINST **19**, T11002 (2024), arXiv:2407.10716.
- [24] B. Ary Dos Santos Garcia et al. (MADMAX Collaboration), *First search for axion dark matter with a MADMAX prototype*, Phys. Rev. Lett. **135**, 041001 (2025), arXiv:2409.11777.
- [25] J. Egge, *Axion haloscope signal power from reciprocity*, JCAP **04** (2023) 064, arXiv:2211.11503.
- [26] J. Egge et al., *Experimental determination of axion signal power of dish antennas and dielectric haloscopes using the reciprocity approach*, JCAP **04** (2024) 005, arXiv:2311.13359.
- [27] L. Di Luzio, M. Giannotti, E. Nardi, L. Visinelli, *The landscape of QCD axion models*, Phys. Rept. **870**, 1 (2020), arXiv:2003.01100.
- [28] S. Antusch, J. Kersten, M. Lindner, M. Ratz, M. A. Schmidt, *Running neutrino mass parameters in see-saw scenarios*, JHEP **03** (2005) 024, arXiv:hep-ph/0501272.
- [29] D. Budker, P. W. Graham, M. Ledbetter, S. Rajendran, A. O. Sushkov, *Proposal for a Cosmic Axion Spin Precession Experiment (CASPEr)*, Phys. Rev. X **4**, 021030 (2014), arXiv:1306.6089.
- [30] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, A. Zhou, *NuFIT 5.2 — Three-neutrino fit of oscillation data*, <http://www.nu-fit.org> (2022).